RAJALAKSHMI ENGINEERING COLLEGE

ABINAVVA 251001504-2025-IT

A2

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|---|---|
| **Started on** | Monday, 28 September 2026, 5:18 PM |
| **State** | Finished |
| **Completed on** | Friday, 9 October 2026, 11:04 AM |
| **Time taken** | 10 days 17 hours |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

**For example:**

| Input | Result |
|---|---|
| 5<br>1 1 2 3 4 | 1 |

**Answer:** (penalty regime: 0 %)

```
#include <stdio.h>

int main(){
    int n;
    scanf("%d",&n);
    int a[n];

    for(int i=0;i<n;i++) scanf("%d",&a[i]);

    int slow=a[0], fast=a[0];

    do{
        slow=a[slow];
        fast=a[a[fast]];
    }while(slow!=fast);

    slow=a[0];

    while(slow!=fast){
        slow=a[slow];
        fast=a[fast];
    }

    printf("%d",slow);
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 11<br>10 9 7 6 5 1 2 3 8 4 7 | 7 | 7 | ✔ |
| ✔ | 5<br>1 2 3 4 4 | 4 | 4 | ✔ |
| ✔ | 5<br>1 1 2 3 4 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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